Exact finite-target seeds and quotient evaluation for a Coxeter search

Bounded L033 method component

Abstract

We replace two operations in the retained LINS normal-subgroup search: finite-target epimorphism enumeration and evaluation of a composed homomorphism. Both changes preserve the complete set of normal kernels. No new hyperbolic quotient, minimum, named group, or completed target bound is asserted by this component.

1 The finite-target list

Let

$$W = \langle s_1, \ldots, s_r \mid s_i^2 = 1, (s_i s_j)^{m_{ij}} = 1 \ (i < j) \rangle$$

for a specified finite Coxeter matrix. Our interface checks the literal presentation, including every relator, against this matrix. Fix a finite target group Q. Every homomorphism $W \to Q$ corresponds to a tuple $(a_1, \ldots, a_r)$ with $a_i^2 = 1$ and $(a_i a_j)^{m_{ij}} = 1$. Identities and proper divisors of the prescribed orders are allowed at this stage. Surjectivity is exactly $\langle a_1, \ldots, a_r \rangle = Q$.

After choosing a prefix $(a_1, \ldots, a_{k-1})$, its stabilizer under simultaneous inner conjugation is $C_Q(a_1, \ldots, a_{k-1})$. The possible next elements are exactly the square-one elements satisfying the relations to all chosen marks. This set is invariant under that stabilizer. Choose one representative from each of its actual conjugation orbits and continue with the centralizer of the extended prefix. Induction proves that the resulting full tuples represent every inner-conjugacy orbit exactly once. Filtering the full tuples by generation and constructing the actual homomorphism therefore finds every epimorphism up to inner conjugation. Inner-conjugate homomorphisms have the same kernel. Outer-conjugate ones may still have the same kernel; the normal-subgroup graph removes these duplicates. No quotient-isomorphism shortcut is used to remove distinct kernels.

The retained default LINS routine first seeds its graph with all kernels of epimorphisms to the finite targets in its table through the bound. Its installed `findTQuotients.gi` explicitly uses `GQuotients` for this operation. Our initialization inserts precisely the same set of kernels through the existing graph-insertion interface. It then disables only that already completed target-seeding pass. All subsequent prime-power quotient, intersection, and graph operations are unchanged, except the evaluation below. The invocation uses the default options apart from this initialization and target-list filter; no cutting or early termination option is supplied. The caller retains the original index guard. The LINS algorithm and its finite classification tables, through their installed 10^7 limit, remain explicit computational and classification premises, as in the accepted eight-diagram core. We do not reprove their completeness.

2 Evaluation in the actual quotient

At a prime-power step LINS constructs an isomorphism $\iota : H \to \widehat{H}$, where $\widehat{H}$ is finitely presented, and an epimorphism $\mu : \widehat{H} \to E$ onto an elementary abelian group. It obtains the conjugation

matrix by evaluating words $w = xyx^{-1} \in H$ as $\mu(\iota(w))$. Set $\nu = \mu \circ \iota$. Then $\nu(w) = \mu(\iota(w))$ for every such word. Consequently every matrix entry, every invariant submodule, every pullback kernel, and every recursive search remains identical.

The process-local override adds the local variable `Nu`, sets `Nu:=Iso*Mu` in GAP's composition notation, and replaces only that nested image by `Image(Nu,w)`. All other executable statements are literal copies of the pinned installed function. No installed package is edited. GAP's `ghomfp.gi` composition method copies the augmented coset table and maps its generator images through the second map. Rewriting can thus evaluate directly in E instead of building a long intermediate word in $\widehat{H}$. This implementation explains the possible saving; the mathematical equality above does not depend on a timing assumption or an injectivity claim for μ.

Controls and scope. A literal traversal of all square-one tuples independently agrees with the inner-orbit traversal for three matrices and six finite groups (18 cases, seven surjective inner orbits). Complete old and new normal kernel inventories agree bijectively as ordered marked quotients for three bounded presentations. During those changed searches all 676 composed conjugation images are also evaluated by the original nested formula and compared exactly. Invalid matrices and mismatched source presentations are rejected. The independent core checker still requires full intersection and actual flag-transitivity tests for positive hypertopes; this method does not weaken them.

The preserved first control driver accidentally captured the replacement function as its reference. Its result is not used for the original-versus-new traversal comparison; the corrected driver reruns all controls while capturing the installed function first. A pair of capped diagnostic profiles shows one index-1920, dimension-71 matrix step taking about 12.7 seconds by nested evaluation and 1.1 seconds by composition. Neither profile completes its target bound, and neither is a new census or general performance theorem. All earlier capped searches remain uncredited.

Attribution. The finite targets and surrounding algorithm are the installed LINS 0.9 implementation, based on Firth's normal-subgroup algorithm; the direct composition method is already present in GAP 4.15.1. Stabilizer-orbit traversal is standard. The present author supplies the Coxeter interface, exact substitution proof, code, and bounded controls. The root supplied the original L033 probe; a separate reader supplied the acquired LINS interface audit. No novelty, whole-source, measurement, question, or Lean credit is claimed.